Lab 5, Act 2: finding the peak when calculus is not practical!

Highway noise vs. frog chorus

NRES 746

Answer key. Answers are in magenta.

The scenario

Males of the frog species from Act 1 call to attract mates, and traffic noise can interfere with frog choruses. A highway runs through the spring complex. To see how far the road's effect reaches, you visit 15 ponds at different distances from the road and count the calling males you hear during one 5-minute listen at each pond. Here are your observed measurements:

Pond	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Distance (km)	0.1	0.2	0.3	0.5	0.7	1	1.2	1.5	2	2.5	3	4	5	6	8
Males heard	1	0	1	2	6	4	8	11	13	5	12	6	11	12	14

A reasonable model is that the expected count rises with distance and levels off far from the road:

$$\lambda(d) = a\left(1 - e^{-bd}\right)$$

This is the monomolecular function from Bolker's bestiary (Ch. 3). Bolker's worked examples in Ch. 6 (p. 243-249) are a useful guide.

How far from the road does the chorus recover, and how strong is the evidence that the road matters at all?

Try Part 1 on paper first, before you open a laptop or start typing code.

Part 1: Plan it on paper (individual, ~25 min)

1a. What do a and b mean biologically, and what are their units?

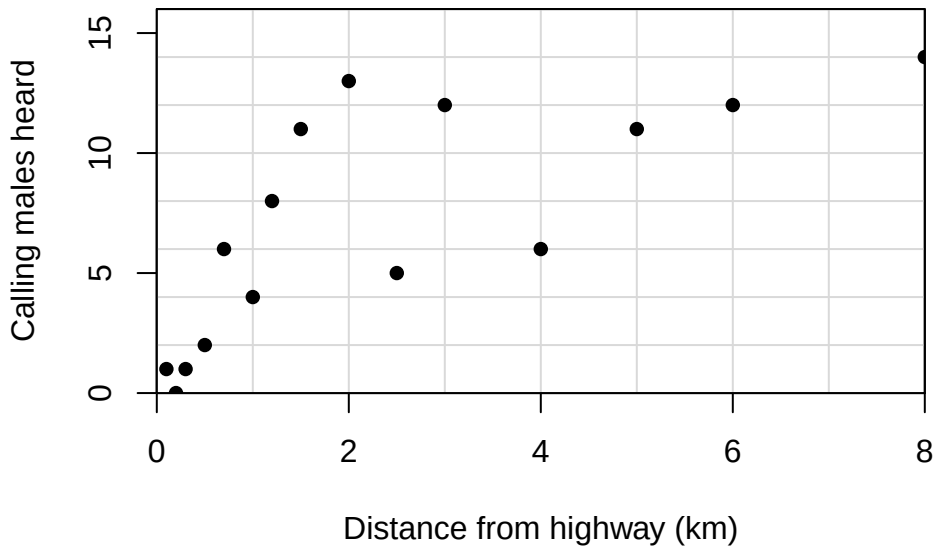
a: the expected number of males heard far from the road, where the chorus has fully recovered (males per 5-minute count). b: how fast the chorus recovers with distance (1/km); larger b means faster recovery.

1b. Act 1 used a binomial distribution. Why is a Poisson distribution a more reasonable choice for these counts?

The counts have no fixed number of trials or upper limit: we don't know how many males are present, only how many we heard. The binomial needs a known N.

1c. Use the figure below and your function to estimate starting values for a and b. Show how you got your estimated starting value for b.

Hint for b: from the figure, estimate the distance $d_{1/2}$ at which the chorus is halfway to its maximum, so that $\lambda(d_{1/2}) = a/2$. Then solve that equation for b. Bolker's bestiary of functions (Ch. 3) shows how parameters like these relate to features of the curve.



$a \approx 12$ (where the points level off). Half of a (about 6) is reached near $d_{1/2} \approx 0.7$ to 1 km. $a(1 - e^{-bd_{1/2}}) = a/2 \Rightarrow e^{-bd_{1/2}} = 1/2 \Rightarrow b = \ln 2/d_{1/2}$.

$a \approx 12$ (11 to 13 is fine) $\quad b \approx \ln 2/0.85 \approx 0.8$ (0.7 to 1 is fine)

1d. Write the negative log-likelihood, NLL(a, b), for the 15 ponds using the Poisson pmf. Simplify it as far as you can. As in Act 1 (1c), you can drop any term that doesn't depend on a or b, since it won't change where the minimum is.

$$\text{NLL}(a,b) = -\sum_{i=1}^{15} \left[y_i \log \lambda_i - \lambda_i - \log(y_i!) \right] \longrightarrow \sum_{i=1}^{15} \left[\lambda_i - y_i \log \lambda_i \right], \qquad \lambda_i = a\left(1 - e^{-bd_i}\right)$$

In Act 1, calculus gave a formula for $\hat{p}$. That rarely happens with real models: for this one, no algebra trick produces a formula for $\hat{a}$ and $\hat{b}$. In practice we usually let a computer search for the minimum numerically instead!

1e. In plain English (not R code), describe step by step what a function must do to compute the NLL for one candidate parameter set (a, b). Then describe how you would use that function to find the MLE.

For a candidate (a, b): compute each pond's expected count λ_i from its distance; compute the log Poisson probability of each observed count given λ_i; add them up; return the negative of the sum. To find the MLE, give this function and a starting guess (from 1c) to an optimizer such as `optim()`, which moves toward lower NLL until it can't improve (a grid search also works).

Checkpoint: finalize your plan (~5 min)

Look back over 1c-1e and fix anything that no longer looks right before you start coding.

Part 2: Fit the model (laptop, ~50 min)

Each coding question is labeled with a recommendation:

- **Recommended: no AI:** try to build this yourself. R's help pages, web search, and classmates are fine.
- **AI optional/encouraged:** I'd recommend trying to solve this with an AI assistant. This implies the results are important but I don't think you absolutely need to know how to code it!

After each one, circle how you actually used AI. Using AI never costs points (but please disclose appropriately!).

What to turn in: your filled-in answers and your 2c figure as an image (upload in webcampus). You don't need to submit your code.

The data are in `FrogChorus.csv` (columns `pond`, `dist_km`, `males`), or you can paste them into R directly:

```
dist_km <- c(0.1, 0.2, 0.3, 0.5, 0.7, 1, 1.2, 1.5, 2, 2.5, 3, 4, 5, 6, 8)
males   <- c(1, 0, 1, 2, 6, 4, 8, 11, 13, 5, 12, 6, 11, 12, 14)
```

2a. *(Recommended: no AI.)* Write a function `nll_mono(par, y, d)` that implements your NLL from 1d, where `par` holds a and b. Check it: `nll_mono(c(10, 1), males, dist_km)` should return about 33.84 if you used `dpois()`, or about -121.31 if you coded the formula yourself and dropped the constant term. (Why can the second number be negative?)

Your value: 33.84 (with `dpois()`), or -121.31 (constant dropped; it can be negative because it's no longer the log of a probability)

AI use (circle one): none / help or debugging / wrote the code

2b. *(recommended: no AI.)* Evaluate the NLL at your starting values from 1c. Then use `optim()`, starting from those values, to find the MLE.

NLL at your starting values: e.g. 33.52 at $(12, 0.8)$

$\hat{a} =$ 12.00 $\hat{b} =$ 0.64 NLL at the MLE: 32.96 (or -122.18 without the constant) convergence code: 0

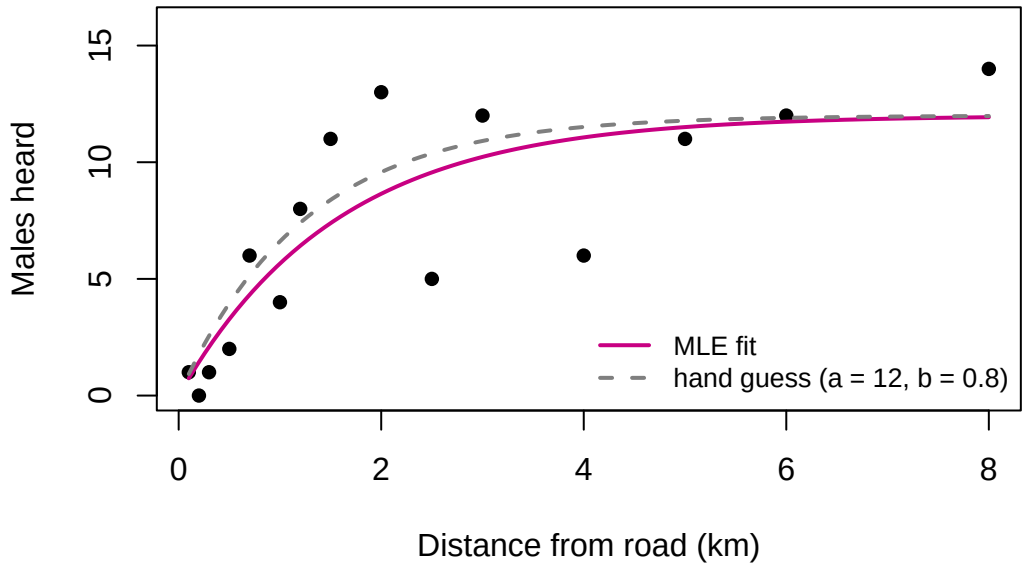
Did `optim()` print any warnings? If so, what do you think caused them, and do they mean the answer is wrong?

Usually "NaNs produced": while searching, `optim()` tries negative a or b, which makes λ_i negative and `dpois()` returns NaN. The answer isn't wrong: convergence code 0, and the fit matches the data. (Fixes: bounds with `"L-BFGS-B"`, or fit $\log a$ and $\log b$.)

AI use (circle one): none / help or debugging / wrote the code

2c. *(AI optional/encouraged.)* Plot the data with your fitted curve and your hand-guessed curve from 1c. How far from the road does the expected chorus reach half its full size?

Half-size distance: $\ln 2/\hat{b} \approx 1.09$ km



AI use (circle one): none / help or debugging / wrote the code

2d. *(recommended: no AI.)* A skeptic says the road has no effect: every pond has the same expected count λ, whatever its distance. What is the MLE of λ under that model, and what is its NLL? (Keep or drop the constant term the same way you did in 2a, so the two NLLs are comparable.) Which model do the data support more? Does the size of the difference convince you?

$\hat{\lambda} = \bar{y} = 106/15 \approx 7.07$ NLL: 53.87 (or -101.27 without the constant)

The road model: its NLL is lower by about 20.9 units, a large difference (the likelihood is about e^{21} times higher), so yes, convincing.

AI use (circle one): none / help or debugging / wrote the code

2e. Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Open-ended.

If you finish early. *(AI optional/encouraged.)* A colleague prefers a different saturating curve, $\lambda(d) = \dfrac{ad}{b+d}$. Fit it the same way and compare its NLL with the monomolecular model's. What do you conclude?

Michaelis-Menten NLL 33.19 vs. 32.96: a difference of only about 0.22. These data can't tell the two curves apart.

AI use (circle one): none / help or debugging / wrote the code

Reflection (~10 min)

In Act 1, a single 5-minute visit detected this frog only about 40% of the time, even at sites where it was known to be present. With that in mind, what do the counts in this Act measure? Could the road lower the counts near it for reasons other than fewer frogs living there? Could these data tell those explanations apart?

Also record any changes you made, new insights, and any "muddiest points" or topics you want to review.

The counts measure males *heard*, not males present. The road could lower counts through fewer frogs, males calling less, or noise masking calls so they're missed. One listen per pond can't separate these.